آمار و احتمال مهندسی

Engineering Probability & Statistics

Lecture Notes ▪ جزوه درسی

Module 6

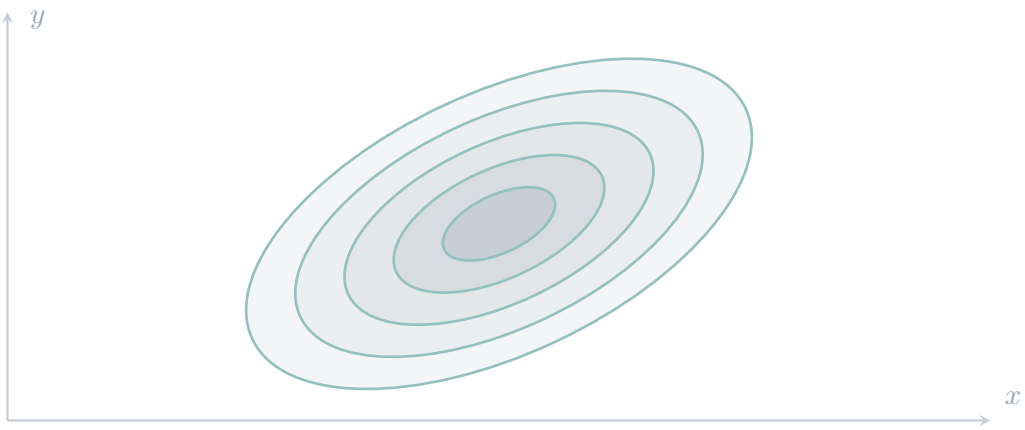
Two Random Variables

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Learning Objectives

After completing this module, students will be able to:

1. Define and compute joint PMF, PDF, and CDF
2. Derive marginal distributions from joint distributions
3. Compute conditional distributions
4. Test independence of two random variables
5. Compute conditional expectations
6. Visualize and analyze joint distributions

6.1 Introduction: Why Study Joint Distributions?

Real engineering systems involve **multiple interacting uncertain quantities**:

- Voltage AND current in a noisy circuit
- Signal AND noise at a receiver
- Temperature AND resistance of a sensor
- I-channel AND Q-channel of a communication signal

Understanding the **joint behavior** tells us things marginals cannot — like correlation, dependence, and conditional behavior.

6.2 Joint PMF (Discrete Case)

Definition

For discrete RVs X, Y:

$$p_{X,Y}(x,y) = P(X = x, Y = y)$$

Properties

1. $p_{X,Y}(x,\, y) \geq 0$
2. $\sum_x \Sigma_y p_{X,Y}(x,\, y) = 1$

Visualization: Probability Table

Example: X = number of retransmissions $(0,\, 1,\, 2)$, Y = acknowledgment delay category $(1 = \text{fast},\, 2 = \text{slow})$

	$Y = 1$	$Y = 2$	$p_X(x)$
$X = 0$	0.30	0.10	0.40
$X = 1$	0.15	0.20	0.35
$X = 2$	0.05	0.20	0.25
$p_Y(y)$	0.50	0.50	1.00

6.3 Joint PDF (Continuous Case)

Definition

For continuous RVs X, Y, the joint PDF $f_{X,Y}(x,\, y)$ satisfies:

$$P((X,Y) \in A) = \iint_A f_{X,Y}(x,y)\, dx\, dy$$

Properties

1. $f_{X,Y}(x,\, y) \geq 0$
2. $\int\int f_{X,Y}(x,\, y)\, dx\, dy = 1$

Visualization

The joint PDF is a **surface** over the $(x,\, y)$ plane. Probability = volume under surface over a region.

Engineering Example: Bivariate Gaussian

Two correlated noise voltages $(X,\, Y)$ with correlation ρ:

$$f_{X,Y}(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\frac{x^2}{\sigma_X^2} - \frac{2\rho xy}{\sigma_X\sigma_Y} + \frac{y^2}{\sigma_Y^2}\right]\right)$$

6.4 Joint CDF

Definition

$$F_{X,Y}(x,y) = P(X \leq x, Y \leq y)$$

Properties

1. $F_{X,Y}(-\infty,\, y) = 0,\; F_{X,Y}(x,\, -\infty) = 0$
2. $F_{X,Y}(\infty,\, \infty) = 1$
3. Non-decreasing in both arguments
4. $f_{X,Y}(x,\, y) = \dfrac{\partial^2 F_{X,Y}}{\partial x \partial y}$

6.5 Marginal Distributions

Obtaining Marginals from Joint

Discrete:

$$p_X(x) = \sum_y p_{X,Y}(x,y), \quad p_Y(y) = \sum_x p_{X,Y}(x,y)$$

Continuous:

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy, \quad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dx$$

Visual Interpretation

The marginal PDF $f_X(x)$ is the **projection** (integral) of the joint surface onto the x-axis. Similarly $f_Y(y)$ projects onto the y-axis.

Engineering Example

Joint PDF: $f_{X,Y}(x,\,y) = 2e^{-x}e^{-2y}$, $x \geq 0$, $y \geq 0$
Marginals:

- $f_X(x) = \displaystyle\int_0^{\infty} 2e^{-x}e^{-2y}\,dy = 2e^{-x} \cdot \left[\frac{1}{2}\right] = e^{-x} \to X \sim \text{Exp}(1)$
- $f_Y(y) = \int_0^{\infty} 2e^{-x}e^{-2y}\,dx = 2e^{-2y} \cdot [1] = 2e^{-2y} \to Y \sim \text{Exp}(2)$

6.6 Conditional Distributions

Discrete

$$p_{X|Y}(x|y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}$$

Continuous

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

Interpretation

The conditional distribution describes X when we **know** $Y = y$. It's a "slice" of the joint distribution at a fixed y value, renormalized to integrate to 1.

Engineering Example: Signal Given Channel State

Received power Y given channel state X:

- If channel is good ($X = 1$): $Y \sim N(10,\ 1)$
- If channel is poor ($X = 0$): $Y \sim N(2,\ 4)$
- $P(X = 1) = 0.7$, $P(X = 0) = 0.3$

Conditional PDFs describe receiver behavior in each state.

6.7 Independence of Two Random Variables

Definition

X and Y are independent if and only if:

$$f_{X,Y}(x,y) = f_X(x) \cdot f_Y(y) \quad \text{for all } x, y$$

(Or equivalently for PMFs: $p_{X,Y}(x,\, y) = p_X(x) \cdot p_Y(y)$)

Consequences of Independence

- $f_{X|Y}(x \mid y) = f_X(x)$ (knowing Y doesn't change X)
- $E[XY] = E[X]E[Y]$
- $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$

Testing Independence

Check: Does the joint factor into a product of a function of x alone and y alone?

- $f_{X,Y}(x,\, y) = 2e^{-x} \cdot e^{-2y} = [e^{-x}] \cdot [2e^{-2y}]$ ✔ Independent!
- $f_{X,Y}(x,\, y) = x + y$ for $0 \le x,\, y \le 1 \rightarrow$ cannot factor $\rightarrow$ NOT independent

Engineering Example: Signal and Independent Noise

Transmitted signal S and channel noise N are independent by physical assumption:
$f_{S,N}(s,\, n) = f_S(s) \cdot f_N(n)$
This independence assumption is fundamental to communication system design.

6.8 Conditional Expectation

Definition

$$E[X|Y=y] = \begin{cases} \sum_x x \cdot p_{X|Y}(x|y) & \text{discrete} \\ \int_{-\infty}^{\infty} x \cdot f_{X|Y}(x|y)\,dx & \text{continuous} \end{cases}$$

Interpretation

$E[X \mid Y = y]$ is the **best prediction** of X given that we observed $Y = y$ (in the mean-square sense).

Engineering Example: Temperature-Dependent Resistance

A resistor's value R depends on temperature T. If $R \mid T \sim N(1000 + 0.5T,\, 4)$:
$E[R \mid T = 50°C] = 1000 + 0.5(50) = 1025\,\Omega$
Given temperature, we can predict resistance.

6.9 Practice Problems

Problem 1

Joint PDF: $f_{X,Y}(x,\,y) = 6(1-y)$ for $0 \le x \le y \le 1$, zero elsewhere.

(a) Verify normalization. (b) Find marginals. (c) Are X,Y independent? (d) Find $P(X < 0.5,\, Y < 0.5)$.

Solution

(a) $\int_0^1 \int_0^y 6(1-y)\,dx\,dy = \int_0^1 6y(1-y)\,dy = 6\left[\frac{y^2}{2} - \frac{y^3}{3}\right]_0^1 = 6\left(\frac{1}{2} - \frac{1}{3}\right) = 1$ ✔

(b) $f_X(x) = \int_x^1 6(1-y)\,dy = 6\left[\left(1 - \frac{y^2}{2}\right) - (y)\right]_x^1 = 3(1-x)^2$; $f_Y(y) = 6y(1-y)$

(c) $f_{X,Y} \neq f_X \cdot f_Y \to$ NOT independent (also, the support is triangular, not rectangular)

(d) $P(X < 0.5,\, Y < 0.5) = \displaystyle\int_0^{0.5}\int_0^{y} 6(1-y)\,dx\,dy = \int_0^{0.5} 6y(1-y)\,dy = 6\left[\frac{y^2}{2} - \frac{y^3}{3}\right]_0^{0.5} = 0.500$

Problem 2

Two sensors measure the same signal with independent noise. $X = S + N_1$, $Y = S + N_2$ where $S = 5\,\text{V}$, $N_1 \sim N(0,\,1)$, $N_2 \sim N(0,\,4)$. Are X and Y independent? Find $E[X]$, $E[Y]$, and explain why $\text{Cov}(X,\,Y) \neq 0$.

Solution X and Y are NOT independent because they share S (though noise is independent). $E[X] = E[Y] = 5$. $\text{Cov}(X,\,Y) = \text{Cov}(S + N_1,\, S + N_2) = \text{Var}(S) + \text{Cov}(S,\, N_1) + \text{Cov}(N_2,\, S) + \text{Cov}(N_1,\, N_2)$. Since S is constant, $\text{Var}(S) = 0$, so if S were random they'd be correlated. With S fixed, X and Y are actually independent in this case. If S is random with $\text{Var}(S) = \sigma_S^2$, then $\text{Cov}(X,\,Y) = \sigma_S^2$.

Problem 3

Given joint PMF in the table of Section 6.2, find (a) $P(X \le 1,\, Y = 2)$, (b) $f_{X|Y}(x \mid Y = 1)$, (c) $E[X \mid Y = 1]$.

Solution

(a) $P(X \le 1,\, Y = 2) = p(0,\,2) + p(1,\,2) = 0.10 + 0.20 = 0.30$

(b) $p_{X|Y}(x \mid 1) = \frac{p(x,\,1)}{p_Y(1)}$: $p(0 \mid 1) = \frac{0.30}{0.50} = 0.60$, $p(1 \mid 1) = \frac{0.15}{0.50} = 0.30$, $p(2 \mid 1) = \frac{0.05}{0.50} = 0.10$

(c) $E[X \mid Y = 1] = 0(0.60) + 1(0.30) + 2(0.10) = 0.50$

Next Module: Module 7 — Functions of Two Random Variables